Shell Scripting Machine Test – Mini Projects

Project 1 – Backup Automation Script

Requirements:
1. Create a backup folder if it does not exist.
2. Take source folder input from the user.
3. Verify that the source folder exists.
4. Copy all files to the backup folder.
5. Create a timestamped compressed archive (.tar.gz).
6. Generate a backup log file.
7. Store date and time in logs.
8. Display success/failure messages.
9. Use variables, conditions, functions, and loops.
10. Schedule the script using Cron.

Project 2 – User Management Automation Script

Requirements:
1. Ask the user how many Linux users to create.
2. Accept usernames using a loop.
3. Check whether each user already exists.
4. Create new users automatically.
5. Skip existing users.
6. Generate user creation logs.
7. Store username, date, and time.
8. Use variables, conditions, functions, and loops.
9. Display appropriate status messages.
10. Verify successful execution.

Project 3 – Disk Space Monitoring & Alert Automation Script

Requirements:
1. Check disk usage automatically.
2. Display current disk usage percentage.
3. Generate alerts when usage exceeds a threshold.
4. Create monitoring logs.
5. Store date, time, and usage details.
6. Use variables, conditions, functions, and loops.
7. Create separate ALERT and NORMAL log entries.
8. Display monitoring status.
9. Schedule execution every minute using Cron.
10. Verify log generation.

Project 4 – Log File Cleanup Automation Script

Requirements:
1. Ask the user for a log folder path.
2. Ask for the number of days.
3. Verify the folder exists.
4. Identify old .log files.
5. Display files that will be deleted.

6. Remove old log files.
7. Generate cleanup logs.
8. Store deleted file names, date, and time.
9. Use variables, conditions, functions, and loops.
10. Schedule the script using Cron.

Project 5 – Memory Usage Monitoring Automation Script

Requirements:
1. Display total, used, and free memory.
2. Ask the user for an alert threshold.
3. Compare memory usage with the threshold.
4. Generate ALERT messages when exceeded.
5. Generate NORMAL logs otherwise.
6. Store date, time, and memory usage.
7. Create a separate logs folder.
8. Use variables, conditions, functions, and loops.
9. Display monitoring status.
10. Schedule execution every minute using Cron.

Project 6 – Directory File Count Monitoring Script

Requirements:
1. Ask the user for a directory path.
2. Verify the directory exists.
3. Count total files in the directory.
4. Display the file count.
5. Ask for a maximum file limit.
6. Compare count with the limit.
7. Generate alerts when the limit is exceeded.
8. Generate normal logs otherwise.
9. Store date, time, directory name, and file count.
10. Use variables, conditions, functions, and loops.
11. Generate log files automatically.
12. Schedule execution using Cron.